

PROGRAM:

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#include <stdio.h>
#include <stdlib.h>
#define MAX_VERTICES 100
struct Node {
    int vertex;
    struct Node* next;
};
struct Graph {
    struct Node* adjLists[MAX_VERTICES];
    int visited[MAX_VERTICES];
};
// Create a node
struct Node* createNode(int v) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->vertex = v;
    newNode->next = NULL;
    return newNode;
}
// Create a graph
struct Graph* createGraph() {
    struct Graph* graph = (struct Graph*)malloc(sizeof(struct Graph));
    int i;
    for (i = 0; i < MAX_VERTICES; i++) {
        graph->adjLists[i] = NULL;
        graph->visited[i] = 0;
    }
    return graph;
}
// Add edge
void addEdge(struct Graph* graph, int src, int dest) {
    // Add edge from src to dest
    struct Node* newNode = createNode(dest);
    newNode->next = graph->adjLists[src];
    graph->adjLists[src] = newNode;
    // Add edge from dest to src
    newNode = createNode(src);
    newNode->next = graph->adjLists[dest];
    graph->adjLists[dest] = newNode;
}
// Print the graph
void printGraph(struct Graph* graph) {
    int v;
```

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for (v = 0; v < MAX_VERTICES; v++) {
    struct Node* temp = graph->adjLists[v];
    printf("\n Adjacency list of vertex %d\n ", v);
    while (temp) {
        printf("%d -> ", temp->vertex);
        temp = temp->next;
    }
    printf("\n");
}

// DFS traversal
void DFS(struct Graph* graph, int startVertex) {
    struct Node* adjList = graph->adjLists[startVertex];
    struct Node* temp = adjList;
    graph->visited[startVertex] = 1;
    printf("Visited %d \n", startVertex);
    while (temp != NULL) {
        int connectedVertex = temp->vertex;
        if (graph->visited[connectedVertex] == 0) {
            DFS(graph, connectedVertex);
        }
        temp = temp->next;
    }
}

int main() {
    struct Graph* graph = createGraph();
    addEdge(graph, 0, 1);
    addEdge(graph, 0, 2);
    addEdge(graph, 1, 2);
    addEdge(graph, 1, 3);
    addEdge(graph, 2, 3);
    printf("Graph:\n");
    printGraph(graph);
    printf("\nDFS Traversal starting from vertex 0:\n");
    DFS(graph, 0);
    return 0;
}

```